

PROVISIONAL PATENT SPECIFICATION

Title: Autonomous AI-Based Tender Processing and Submission System

Applicant: Panacea Trust

Inventor: Lechesa Manaba

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International Patent Classification (IPC)

- G06Q 10/00 – Administration and business processing systems
- G06Q 30/00 – Commerce and procurement systems
- G06N 20/00 – Artificial intelligence systems

1. FIELD OF THE INVENTION

This invention relates to artificial intelligence systems applied to procurement automation. More specifically, it relates to systems capable of autonomously identifying, evaluating, pricing, generating, and submitting tender responses without human intervention.

2. BACKGROUND

Conventional tendering processes rely heavily on manual effort, including searching for opportunities, evaluating requirements, preparing pricing, compiling documentation, and submitting responses. These processes are time-consuming, inconsistent, and limit scalability.

3. PROBLEM TO BE SOLVED

There is a need for a system capable of automating the full tender lifecycle, reducing human involvement, improving speed and accuracy, and enabling scalable participation in procurement opportunities.

4. SUMMARY OF THE INVENTION

The invention provides a fully automated system comprising multiple interconnected modules that operate in a pipeline to process tenders from identification through to submission.

5. SYSTEM ARCHITECTURE

The system consists of a modular architecture including data ingestion, filtering, scoring, pricing, document generation, and submission components coordinated by an automation controller.

6. DATA HARVESTING MODULE

This module retrieves tender data from multiple sources such as government portals, databases, and APIs, and standardizes the information for further processing.

7. FILTERING ENGINE

The filtering engine removes non-qualifying tenders based on predefined rules such as category, location, profit thresholds, and exclusion of tenders requiring physical briefing sessions.

8. SCORING ENGINE

The scoring engine evaluates remaining tenders using weighted criteria including profitability, compliance, strategic alignment, and geographic relevance.

9. PRICING ENGINE

The pricing engine calculates cost and applies predefined markups, tax considerations, and supplier data to generate commercially viable quotations.

10. DOCUMENT GENERATION

This module produces structured quotation documents including line items, totals, company details, and compliance information.

11. PDF PACKAGING

The system converts generated documents into standardized PDF formats suitable for submission.

12. SUBMISSION ENGINE

This module submits tender responses via electronic channels such as email or integrated APIs.

13. PROCESS FLOW

The system follows a sequential workflow: data retrieval, filtering, scoring, pricing, document generation, PDF conversion, and submission.

14. DECISION ENGINE

Decision logic is based on weighted scoring models ensuring only high-value opportunities proceed through the pipeline.

15. TECHNICAL IMPLEMENTATION

The system is implemented using APIs, databases, asynchronous processing, and automation frameworks enabling continuous operation.

16. VARIATIONS

The invention may be implemented as a cloud-based system, distributed architecture, or integrated platform without departing from its core functionality.

17. CLAIMS

A computer-implemented system comprising modules for ingestion, filtering, scoring, pricing, document generation, and submission operating autonomously. A method comprising retrieving, evaluating, pricing, generating, and submitting tenders automatically.

18. ADVANTAGES

The system improves efficiency, reduces manual effort, increases scalability, and enhances profitability.

19. INDUSTRIAL APPLICATION

Applicable in government procurement, private sector sourcing, and automated business platforms.

20. ABSTRACT

An artificial intelligence-based system that automates the full tender process from identification to submission.